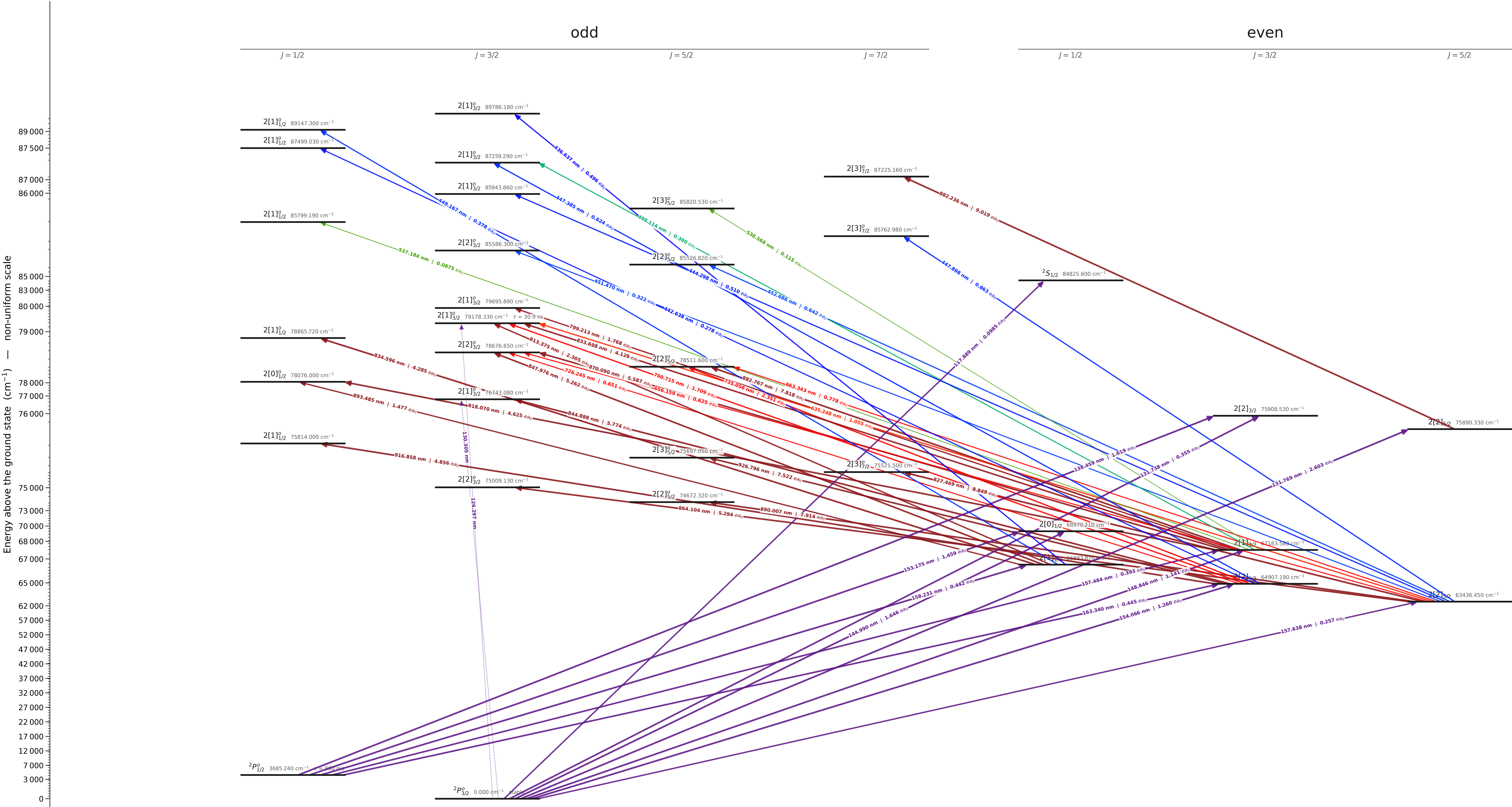
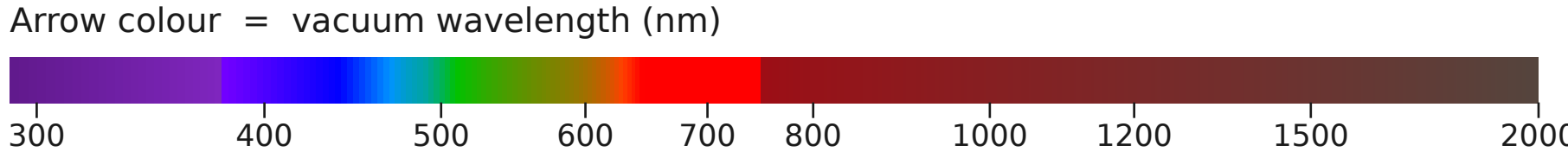


Bromine

Br (Br I). Strongest classified transitions below 2 μm (NIST): 33 levels, 47 lines.

Br⁺ ionisation limit 95284.80 cm⁻¹ = 11.81381 eV (104.949 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $[J||er||J']$ in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model or semi-empirical calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, *n* = 70.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$[J er J']$ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
² P _{3/2} ^o - 2[1] _{3/2}	126.2972	-	2373.7066	-	-	5.151	-	TALIF detection of ground-state Br: 2 x 252.594 nm, fluor...
² P _{3/2} ^o - 2[1] _{3/2}	130.3049	-	2300.6997	-	-	-	-	two-photon LIF scheme near 260.6 nm with fluorescence 5p...
² P _{3/2} ^o - 2[1] _{3/2}	148.8459	-	2014.1131	1.141	2.000e+08	-	-	148.8 nm resonance line
² P _{3/2} ^o - 2[2] _{3/2}	154.0661	-	1945.8686	1.260	2.200e+08	-	-	154.1 nm resonance line
² P _{3/2} ^o - 2[1] _{3/2}	157.4844	-	1903.6323	0.393	2.000e+07	-	-	157.5 nm line to the metastable 2P°1/2 level
² P _{3/2} ^o - 2[2] _{5/2}	157.6381	-	1901.7769	0.257	5.700e+06	-	-	157.6 nm resonance line
² P _{3/2} ^o - 2[2] _{3/2}	163.3401	-	1835.3879	0.445	2.300e+07	-	-	163.3 nm line to the metastable 2P°1/2 level
2[2] _{5/2} - 2[1] _{3/2}	635.2481	635.0725	471.9297	1.055	2.200e+06	5.151	6.8	635.25 nm fluorescence line used to detect Br atoms after...

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

⁷⁹Br hyperfine structure (I = 3/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
² P _{3/2}	884.80977(6)	-384.88284(20)	F=0: -3799.14 F=1: -2529.45 F=2: -374.94 F=3: +1894.60
² P _{1/2}	5332.55(4)	-	F=1: -6665.69 F=2: +3999.41

Measured A, B constants; sources in the data file.